

COVID-19 vaccination around immune checkpoint inhibitor start and survival in a nationwide cohort of patients with cancer

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Experimental data suggest that COVID-19 vaccination could modulate immune checkpoint inhibitor (ICI) effects, but population evidence is limited. Using the French health data system (*Système National des Données de Santé*), we identified 95,015 adults initiating ICI therapy between December 2020 and March 2023. We assessed mRNA vaccination within 100 days before ICI initiation using inverse probability of treatment weighting and within 100 days after ICI initiation among previously unvaccinated patients using a cloning–censoring–weighting target-trial emulation. The primary outcome was all-cause mortality; secondary outcomes were cancer-specific and non-COVID-19 mortality. Follow-up extended to 31 December 2024 for all-cause mortality and to 31 December 2023 for cause-specific deaths. Pre-ICI mRNA vaccination was associated with lower early all-cause mortality (hazard ratio (HR) 0.90 (95% confidence interval (CI) 0.87–0.94) at 1–3 months and 0.96 (0.93–1.00) at 6 months) and no significant association after 12 months. Post-ICI mRNA vaccination showed a similar early association (HR 0.84 (95% CI 0.80–0.88) at 3 months and 0.86 (0.79–0.94) at 6 months), also with no significant association after 12 months. Estimates for non-COVID-19 and cancer-specific mortality were similar to those for overall survival. Similar early patterns with non-mRNA COVID-19 and other adult vaccines suggest modest, transient associations or healthy-vaccine effects, rather than a specific mRNA COVID-19 vaccine effect.

Immune checkpoint inhibitors (ICIs) have transformed the management of many advanced cancers¹, yet only a subset of patients derive durable benefit, and the determinants of ICI real-world effectiveness in routine practice remain incompletely understood^{2,3}. Host-intrinsic factors, such as tumor mutational burden and programmed death-ligand 1 (PD-L1) expression^{2,3}, interact with extrinsic modulators of immunity, including infections and vaccinations^{4,5}, to shape antitumor responses and toxicities. The COVID-19 pandemic created a unique situation in which large numbers of patients with cancer received COVID-19

vaccines in close temporal proximity to ICI initiation, raising questions on both clinical outcomes and the possibility that these vaccines might alter ICI efficacy⁴. mRNA COVID-19 vaccines induce robust innate and adaptive anti-viral immune responses, characterized by type I interferon production and activation of antigen-presenting cells, together with modulation of myeloid cells and the tumor microenvironment compartment, including PD-L1 expression and antitumor immune response^{6–8}. A very recent experimental work has suggested that these properties could ‘sensitize’ otherwise poorly immunogenic tumors to

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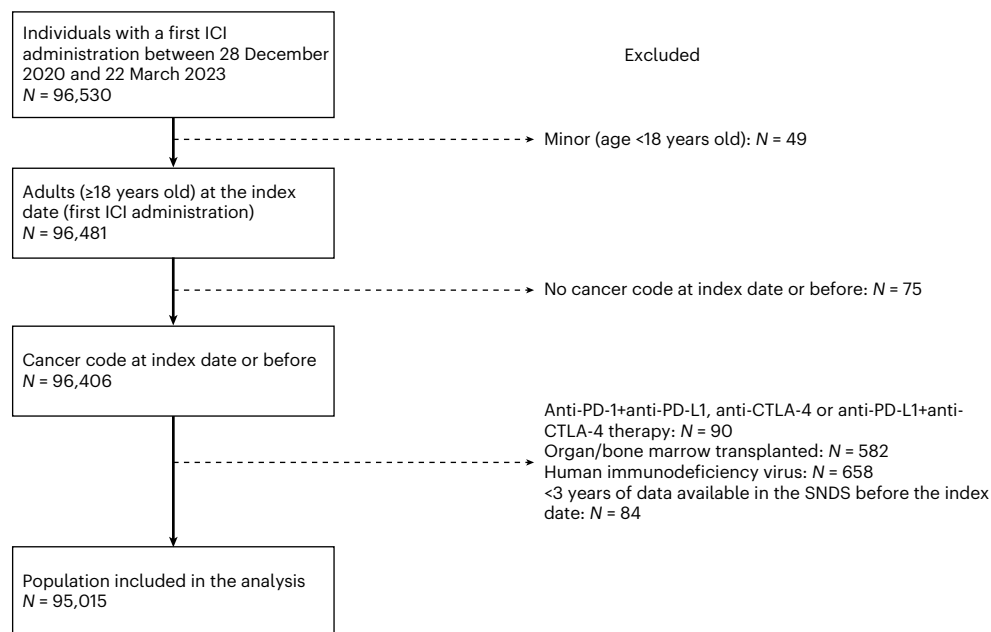


Fig. 1 | Study flowchart.

programmed cell death protein 1 (PD-1)/PD-L1 blockade⁸. In a clinical and mechanistic single-center study, Grippin et al.⁸ reported a marked early survival association among patients with advanced non-small cell lung cancer and melanoma who received COVID-19 mRNA vaccination around ICI administration (within 100 days before or after ICI start), supported by biological data. However, these lung cancer and melanoma cohorts were relatively small (hundreds of patients), and the analysis was subject to several sources of bias, including immortal-time bias, healthy-vaccinee effects⁹ and positivity issues related to calendar time (with inclusion periods extending to times before COVID-19 vaccines became available), underscoring the need for independent confirmation in larger, population-based settings using designs that explicitly address these issues¹⁰.

To date, observational studies of COVID-19 vaccination in ICI-treated patients have primarily focused on safety, immunogenicity or short-term tumor response^{4,5,11–13}, and have generally relied on conventional regression approaches that treat vaccination as a fixed baseline exposure. Consequently, it remains unknown whether peri-ICI COVID-19 vaccination is associated with improved survival at the population level, to what extent any association is explained by protection against COVID-19 itself, rather than by modulation of antitumor immunity, and whether the signal reported in highly selected cohorts can be reproduced across diverse tumor types and care settings.

Here, using the French National Health Data System (*Système National des Données de Santé* (SNDS))^{14,15}, we applied causal inference methods for observational data to the nationwide cohort of all adults initiating ICI therapy between the introduction of COVID-19 vaccines in France and the end of the period during which vaccination was exhaustively recorded in the SNDS. We examined whether receiving COVID-19 vaccination—particularly mRNA vaccination—within a 100-day prespecified window before or after ICI initiation was associated with all-cause mortality, while adjusting for multiple prognostic covariates, using inverse probability of treatment weighting (IPTW) and a cloning–censoring–weighting (CCW) approach, respectively, to account for time-dependent vaccination and immortal-time bias¹⁰. We additionally evaluated cancer-specific and non-COVID-19 mortality. Complementary analyses assessed consistency across major cancer sites, stratified by vaccine platform (mRNA versus non-mRNA), considered documented COVID-19 infection and non-COVID adult vaccinations as

control exposures, and used positive- and negative-control outcomes to explore residual confoundings.

Results

Study population and COVID-19 vaccination patterns

Between 28 December 2020 (first availability of COVID-19 vaccines in France) and 22 March 2023 (end of available COVID-19 infection reporting in France), we included 95,015 adults initiating ICI in the SNDS (flowchart in Fig. 1; Table 1 for general characteristics) with a median (interquartile range (IQR)) follow-up of 32.6 months (26.7–39.6). The median age at ICI initiation was 67 years (60–74), and 65.2% were men. Lung and other respiratory cancers represented nearly half of the cohort (45,825; 48.2%), followed by head and neck (7,216; 7.6%), melanoma (7,147; 7.5%), kidney (6,257; 6.6%), urovesical (6,233; 6.6%), liver (5,495; 5.8%), gastrointestinal (4,372; 4.6%) and breast (4,094; 4.3%), and the rest were in other cancer sites (Fig. 2a). Most patients received anti-PD-1 monotherapy (71.2%), with smaller proportions treated with anti-PD-L1 (23.5%) or anti-PD-1 plus anti-cytotoxic T-lymphocyte-associated protein 4 (anti-CTLA-4) (5.3%; Fig. 2b). Within 3 years of ICI start, 23.9% had major cancer surgery; 49.5% and 33.4% received chemotherapy and radiotherapy, respectively; and 10.1% of patients received anti-angiogenic combined or within 3 months before ICI initiation. Comorbidities were frequent, including hypertension (58.6%), diabetes (23.3%), chronic obstructive pulmonary disease/emphysema (18.5%) and heart failure (7.3%). The details per cancer type are shown in Supplementary Tables 1–7.

The COVID-19 vaccination patterns in the peri-ICI period of interest are shown in the Sankey diagram in Fig. 2c, highlighting the large predominance of mRNA-only COVID-19 vaccination and the rarity of mixed or non-mRNA schedules. Within the 100 days before ICI initiation (termed pre-ICI period), 27,395 patients (28.8%) had received at least 1 dose of an mRNA COVID-19 vaccine only (median doses (IQR), 1 (1–1)), 2,326 (2.4%) had received at least 1 dose of non-mRNA COVID-19 vaccine only (median doses, 1 (1–1)), 68 (0.1%) had received at least 1 dose of an mRNA COVID-19 vaccine and of a non-mRNA COVID-19 vaccine and 68.6% were not vaccinated for COVID-19 (Fig. 2c). Uptake of mRNA vaccination before ICI initiation was between 28% and 33% in the 6 most prevalent cancer subtypes tested and in the patients with metastatic cancer (see Supplementary Tables 1–7 for details). In

Table 1 | Characteristics of the population at index date by vaccination status within 100 days before ICI initiation (pre-ICI period)

Variables	COVID-19 mRNA vaccine (≥1 dose) ^a	Non-mRNA COVID-19 vaccine (≥1 dose) ^a	No COVID-19 vaccine	Total
Number of patients	27,463	2,326	65,226	95,015
Age (median (IQR), years)	69 (61–75)	66 (61–71)	67 (59–74)	67 (60–74)
Sex (male)	18,582 (67.7%)	1,715 (73.7%)	41,648 (63.9%)	61,945 (65.2%)
Low income (yes)	2,025 (7.4%)	167 (7.2%)	6,871 (10.5%)	9,063 (9.5%)
Deprivation index ^b				
1 (least deprived)	4,965 (18.1%)	351 (15.1%)	10,662 (16.3%)	15,978 (16.8%)
2	5,217 (19.0%)	418 (18.0%)	11,860 (18.2%)	17,495 (18.4%)
3	5,802 (21.1%)	516 (22.2%)	14,478 (22.2%)	20,796 (21.9%)
4	5,851 (21.3%)	527 (22.7%)	13,913 (21.3%)	20,291 (21.4%)
5 (most deprived)	5,628 (20.5%)	514 (22.1%)	14,313 (21.9%)	20,455 (21.5%)
Region of residence ^b				
Auvergne-Rhône-Alpes	3,299 (12.0%)	267 (11.5%)	7,678 (11.8%)	11,244 (11.8%)
Bourgogne-Franche-Comté	1,391 (5.1%)	83 (3.6%)	3,105 (4.8%)	4,579 (4.8%)
Bretagne	1,702 (6.2%)	199 (8.6%)	3,635 (5.6%)	5,536 (5.8%)
Centre-Val de Loire	1,131 (4.1%)	111 (4.8%)	2,611 (4.0%)	3,853 (4.1%)
Corse	175 (0.6%)	7 (0.3%)	489 (0.7%)	671 (0.7%)
Grand Est	2,129 (7.8%)	160 (6.9%)	5,256 (8.1%)	7,545 (7.9%)
Outre-mer	207 (0.8%)	5 (0.2%)	941 (1.4%)	1,153 (1.2%)
Hauts-de-France	2,512 (9.1%)	281 (12.1%)	5,917 (9.1%)	8,710 (9.2%)
Île-de-France	3,926 (14.3%)	257 (11.0%)	9,268 (14.2%)	13,451 (14.2%)
Normandie	1,561 (5.7%)	137 (5.9%)	3,655 (5.6%)	5,353 (5.6%)
Nouvelle-Aquitaine	2,988 (10.9%)	268 (11.5%)	6,735 (10.3%)	9,991 (10.5%)
Occitanie	2,631 (9.6%)	256 (11.0%)	6,828 (10.5%)	9,715 (10.2%)
Pays de la Loire	1,568 (5.7%)	171 (7.4%)	3,324 (5.1%)	5,063 (5.3%)
Provence-Alpes-Côte d'Azur	2,243 (8.2%)	124 (5.3%)	5,784 (8.9%)	8,151 (8.6%)
Year of ICI initiation				
2020–2021	13,826 (50.3%)	2,312 (99.4%)	22,724 (34.8%)	38,862 (40.9%)
2022	11,805 (43.0%)	10 (0.4%)	32,834 (50.3%)	44,649 (47.0%)
2023	1,832 (6.7%)	4 (0.2%)	9,668 (14.8%)	11,504 (12.1%)
ICI regimen				
Anti-PD-1	19,304 (70.3%)	1,611 (69.3%)	46,699 (71.6%)	67,614 (71.2%)
Anti-PD-L1	6,605 (24.1%)	596 (25.6%)	15,139 (23.2%)	22,340 (23.5%)
Anti-PD-1/CTLA-4	1,554 (5.7%)	119 (5.1%)	3,388 (5.2%)	5,061 (5.3%)
Cancer type				
Lung/respiratory	13,638 (49.7%)	1,316 (56.6%)	30,871 (47.3%)	45,825 (48.2%)
Head and neck	2,027 (7.4%)	209 (9.0%)	4,980 (7.6%)	7,216 (7.6%)
Melanoma	2,078 (7.6%)	151 (6.5%)	4,918 (7.5%)	7,147 (7.5%)
Kidney	2,014 (7.3%)	186 (8.0%)	4,057 (6.2%)	6,257 (6.6%)
Urovesical	2,013 (7.3%)	91 (3.9%)	4,129 (6.3%)	6,233 (6.6%)
Liver	1,609 (5.9%)	190 (8.2%)	3,696 (5.7%)	5,495 (5.8%)
Gastrointestinal	1,110 (4.0%)	52 (2.2%)	3,210 (4.9%)	4,372 (4.6%)
Breast cancer ^c	776 (2.8%)	15 (0.6%)	3,303 (5.1%)	4,094 (4.3%)
Skin (malignant, other) ^c	546 (2.0%)	44 (1.9%)	1,259 (1.9%)	1,849 (1.9%)
Mesothelioma/pleural	534 (1.9%)	33 (1.4%)	1,176 (1.8%)	1,743 (1.8%)
Other cancer	1,118 (4.0%)	39 (1.7%)	3,627 (5.5%)	4,784 (5.0%)
Metastatic cancer	17,483 (63.7%)	1,479 (63.6%)	41,595 (63.8%)	60,557 (63.7%)

Table 1 (continued) | Characteristics of the population at index date by vaccination status within 100 days before ICI initiation (pre-ICI period)

Variables	COVID-19 mRNA vaccine (≥1 dose) ^a	Non-mRNA COVID-19 vaccine (≥1 dose) ^a	No COVID-19 vaccine	Total
Poor performance status proxy	4,636 (16.9%)	509 (21.9%)	13,192 (20.2%)	18,337 (19.3%)
Volume of inclusion in the center ^b				
High (top 10%)	2,645 (9.6%)	202 (8.7%)	6,335 (9.7%)	9,182 (9.7%)
Intermediate (10%–40%)	8,522 (31.0%)	715 (30.7%)	19,316 (29.6%)	28,553 (30.1%)
Low (bottom 60%)	16,296 (59.3%)	1,409 (60.6%)	39,575 (60.7%)	57,280 (60.3%)
Major cancer surgery ≤3 years	7,101 (25.9%)	476 (20.5%)	15,137 (23.2%)	22,714 (23.9%)
Chemotherapy ≤3 years	13,519 (49.2%)	978 (42%)	32,498 (49.8%)	46,995 (49.5%)
Radiotherapy ≤3 years	9,458 (34.4%)	741 (31.9%)	21,497 (33%)	31,696 (33.4%)
Anti-angiogenic ≤3 months	2,815 (10.3%)	295 (12.7%)	6,459 (9.9%)	9,569 (10.1%)
≥7mg EQ-PRED ≤3 months	11,547 (42%)	976 (42%)	27,279 (41.8%)	39,802 (41.9%)
Opioid ≤3 months	11,603 (42.2%)	1,109 (47.7%)	28,744 (44.1%)	41,456 (43.6%)
Antibiotic ≤3 months	9,710 (35.4%)	803 (34.5%)	23,901 (36.6%)	34,414 (36.2%)
Antiepileptic ≤3 months	1,282 (4.7%)	124 (5.3%)	3,157 (4.8%)	4,563 (4.8%)
Bisphosphonate ≤3 months	247 (0.9%)	17 (0.7%)	534 (0.8%)	798 (0.8%)
Anti-RANKL ≤3 months	636 (2.3%)	39 (1.7%)	1,220 (1.9%)	1,895 (2%)
Proton pump inhibitors ≤3 months	13,530 (49.3%)	1,158 (49.8%)	32,047 (49.1%)	46,735 (49.2%)
Malnutrition ≤6 months	4,526 (16.5%)	389 (16.7%)	13,451 (20.6%)	18,366 (19.3%)
≥2 ambulance use ≤6 months	3,250 (11.8%)	230 (9.9%)	8,719 (13.4%)	12,199 (12.8%)
≥1 emergency consult ≤6 months	9,281 (33.8%)	824 (35.4%)	23,220 (35.6%)	33,325 (35.1%)
≥1 intensive care unit ≤6 months	3,527 (12.8%)	300 (12.9%)	9,155 (14%)	12,982 (13.7%)
≥1 at-home hospitalization ≤6 months	509 (1.9%)	25 (1.1%)	1,477 (2.3%)	2,011 (2.1%)
≥1 rehabilitation/long-term care ≤6 months	1,076 (3.9%)	51 (2.2%)	2,992 (4.6%)	4,119 (4.3%)
≥1 home nursing care ≤6 months	507 (1.8%)	27 (1.2%)	1,584 (2.4%)	2,118 (2.2%)
Home oxygen ≤6 months	1,290 (4.7%)	118 (5.1%)	3,221 (4.9%)	4,629 (4.9%)
≥1 palliative care stay ≤6 months	989 (3.6%)	71 (3.1%)	3,267 (5%)	4,327 (4.6%)
Heart failure	1,996 (7.3%)	159 (6.8%)	4,748 (7.3%)	6,903 (7.3%)
COPD or emphysema	5,052 (18.4%)	490 (21.1%)	12,003 (18.4%)	17,545 (18.5%)
Respiratory failure	3,373 (12.3%)	317 (13.6%)	8,493 (13%)	12,183 (12.8%)
Diabetes	6,539 (23.8%)	648 (27.9%)	14,941 (22.9%)	22,128 (23.3%)
Obesity	2,437 (8.9%)	239 (10.3%)	5,704 (8.7%)	8,380 (8.8%)
Hypertension	16,661 (60.7%)	1,482 (63.7%)	37,561 (57.6%)	55,704 (58.6%)
COVID-19 infection <100 days	814 (3%)	16 (0.7%)	5,449 (8.4%)	6,279 (6.6%)
Any non-COVID-19 vaccine ≤1 year before pre-ICI period	14,450 (52.6%)	1,365 (58.7%)	27,980 (42.9%)	43,795 (46.1%)
GP consultations ≤6 months				
0–3 consultations	9,339 (34.0%)	482 (20.7%)	22,461 (34.4%)	32,282 (34.0%)
3–6 consultations	9,220 (33.6%)	835 (35.9%)	21,095 (32.3%)	31,150 (32.8%)
>6 consultations	8,904 (32.4%)	1,009 (43.4%)	21,670 (33.2%)	31,583 (33.2%)

COPD, chronic obstructive pulmonary disease; EQ-PRED, equivalent prednisone; GP, general practitioner; RANKL, receptor activator of NF-κB ligand. ^aPatients receiving both mRNA and non-mRNA COVID-19 vaccines ($n=68$) within 100 days before ICI start were included in the mRNA COVID-19 vaccine group for descriptive analyses, resulting in an mRNA-only analysis sample of 27,395 patients. For the pre-ICI analyses, patients who received mRNA and non-mRNA or only a non-mRNA COVID-19 vaccine in this time window were excluded ($n=2,394$). For the post-ICI analyses, patients who received any COVID-19 vaccine were excluded ($n=29,789$). ^bMissing data were limited to volume of inclusion in the center (0.3%), deprivation index (1.9%) and region of residence (0.03%). Missing values were imputed using the median for ordinal variables and the mode for categorical variables. ^cSkin (malignant, other) and breast cancers were included in the other group for the analysis.

the analysis investigating the effect of COVID-19 vaccination within 100 days after ICI initiation (termed post-ICI period), we focused on patients who had received no COVID-19 vaccine in the pre-ICI period ($n=65,226$). Among them, 18,135 (27.8%) initiated a COVID-19 vaccine ($n=17,514$ mRNA only, 1 (1–2) doses; $n=602$ non-mRNA, 1 (1–1) dose; and 19 with at least 1 dose of both types) in the post-ICI period. In total,

47,091 received no COVID-19 vaccine in both the pre- and post-ICI periods. Among patients unvaccinated with a COVID-19 vaccine in the pre-ICI period, 2,176 (3.3%) received a pneumococcal vaccine, and 9,856 (15.1%) received at least 1 dose of any non-COVID-19 vaccine. An UpSet plot showing vaccination overlaps in the pre-ICI period is given in Supplementary Fig. 1.

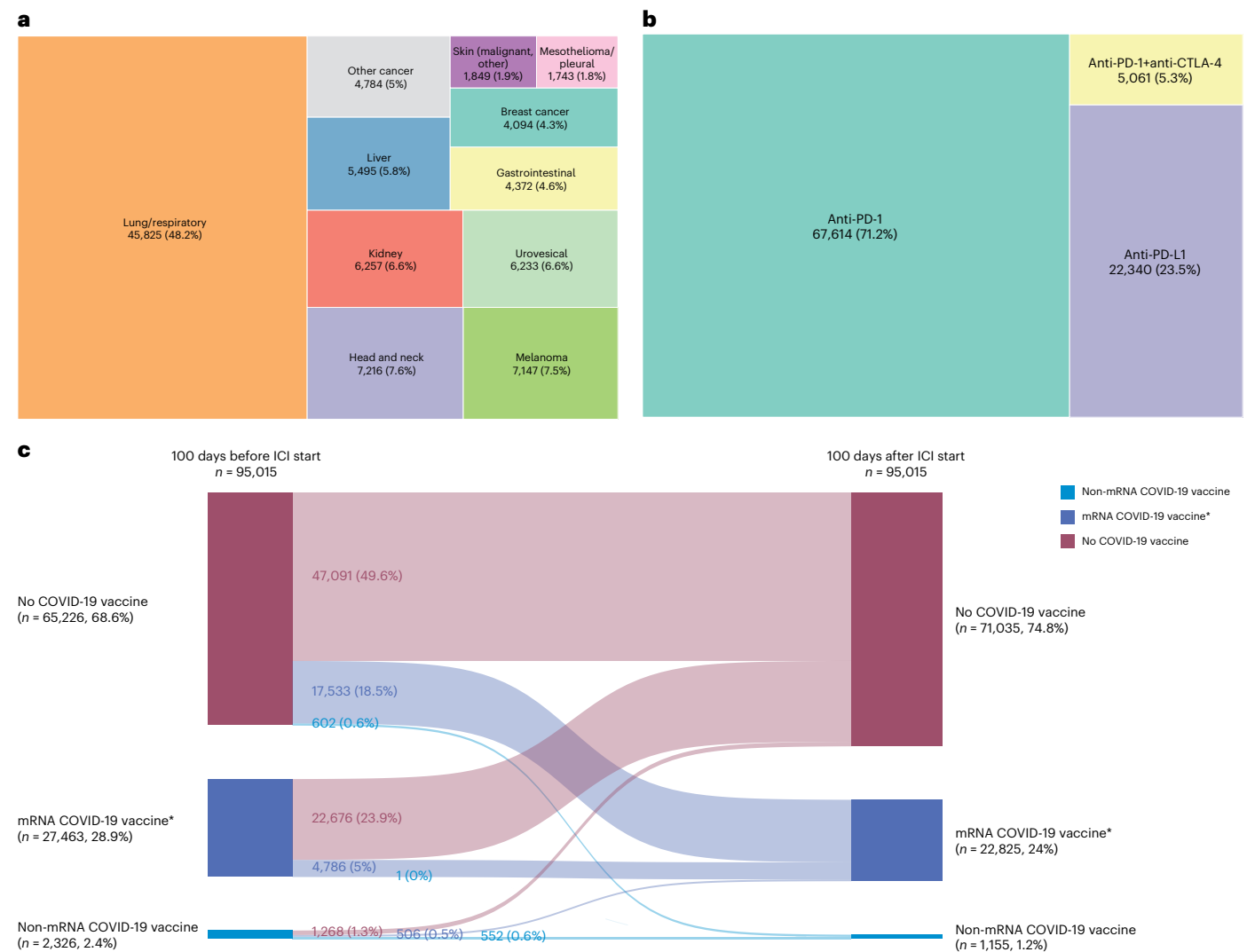


Fig. 2 | Characteristics of the included patients. a–c, Cancer (a), ICI (b), and COVID-19 vaccination (c) profile. *Patients receiving both mRNA and non-mRNA COVID-19 vaccines were included in the mRNA COVID-19 vaccine group for descriptive purposes only ($n = 68$ within 100 days before ICI start and $n = 25$

within 100 days after ICI start). For the primary pre-ICI analysis, patients with mixed mRNA/non-mRNA schedules or only non-mRNA vaccination within the pre-ICI window were excluded, resulting in an mRNA-only analysis sample of 27,395 patients rather than the 27,463 patients shown in the descriptive table.

Pre-ICI COVID-19 vaccination and survival

We first used the cohort of 95,015 adults initiating a first ICI treatment for cancer in France to estimate the effect of having received at least 1 dose of COVID-19 mRNA vaccine ($N = 27,395$), versus no COVID-19 vaccine ($N = 65,226$), in the pre-ICI period on all-cause mortality. Patients who received mRNA and non-mRNA or only a non-mRNA COVID-19 vaccine in this time window were excluded ($N = 2,394$) from the primary analysis. Baseline confounding was addressed using IPTW. After weighting, covariate balance between exposure groups was achieved for all variables, with standardized mean differences below 0.1 (Supplementary Fig. 2).

Weighted Kaplan–Meier curves showed improved overall survival in patients who had received an mRNA COVID-19 vaccine in the pre-ICI period versus those unvaccinated for COVID-19 (Fig. 3a). Weighted median overall survival was 18.2 months (95% confidence interval (CI), 17.7–18.7) in the pre-ICI mRNA-vaccinated group and 17.3 months (95% CI, 17.1–17.6) in the unvaccinated group. At 3 months after ICI start, the estimated survival probability was 86.0% (95% CI, 85.5–86.4) in the pre-ICI COVID-19 mRNA-vaccinated group versus 84.5% (84.2–84.7) in the pre-ICI COVID-19 unvaccinated group, corresponding to an absolute survival difference of 1.5% (1.0–2.0). At 6, 12 and 24 months,

the estimated survival probabilities remained consistently higher in the pre-ICI mRNA-vaccinated group than in the pre-ICI unvaccinated group, with absolute survival differences of 1.5% (0.9–2.1), 1.3% (0.6–2.0) and 1.0% (0.3–1.7), respectively (Supplementary Table 8). In weighted Cox models, the effect of COVID-19 mRNA vaccination in the pre-ICI period on overall survival was time dependent. Hazard ratios (HRs) were therefore estimated at predefined time points after ICI initiation (Fig. 3d), with an HR of 0.9 (0.87–0.94) at 1 and 3 months and then 0.96 (0.93–1.00) at 6 months, and no significant association thereafter with an HR of 1.00 (0.98–1.03). E values (confidence limit closest to the null) were modest for these associations—1.36 (1.26) at 1 and 3 months, and 1.20 (1.00) at 6 months—supporting cautious interpretation given the possibility of residual unmeasured confounding.

Results were consistent across subgroups, in patients with metastatic cancer and main cancer subtypes (lung/respiratory, head and neck, and kidney; Fig. 3d and Supplementary Fig. 4). Sensitivity analysis (Fig. 4) restricting length of pre-ICI period exposure to COVID-19 mRNA vaccine to 50 days (versus 100 days in the main analysis) showed a similar pattern of time-varying beneficial effect on overall survival. Analyses considering overall survival with COVID-19 death as a competing event (Fig. 3b,d) yielded very similar estimates, indicating that

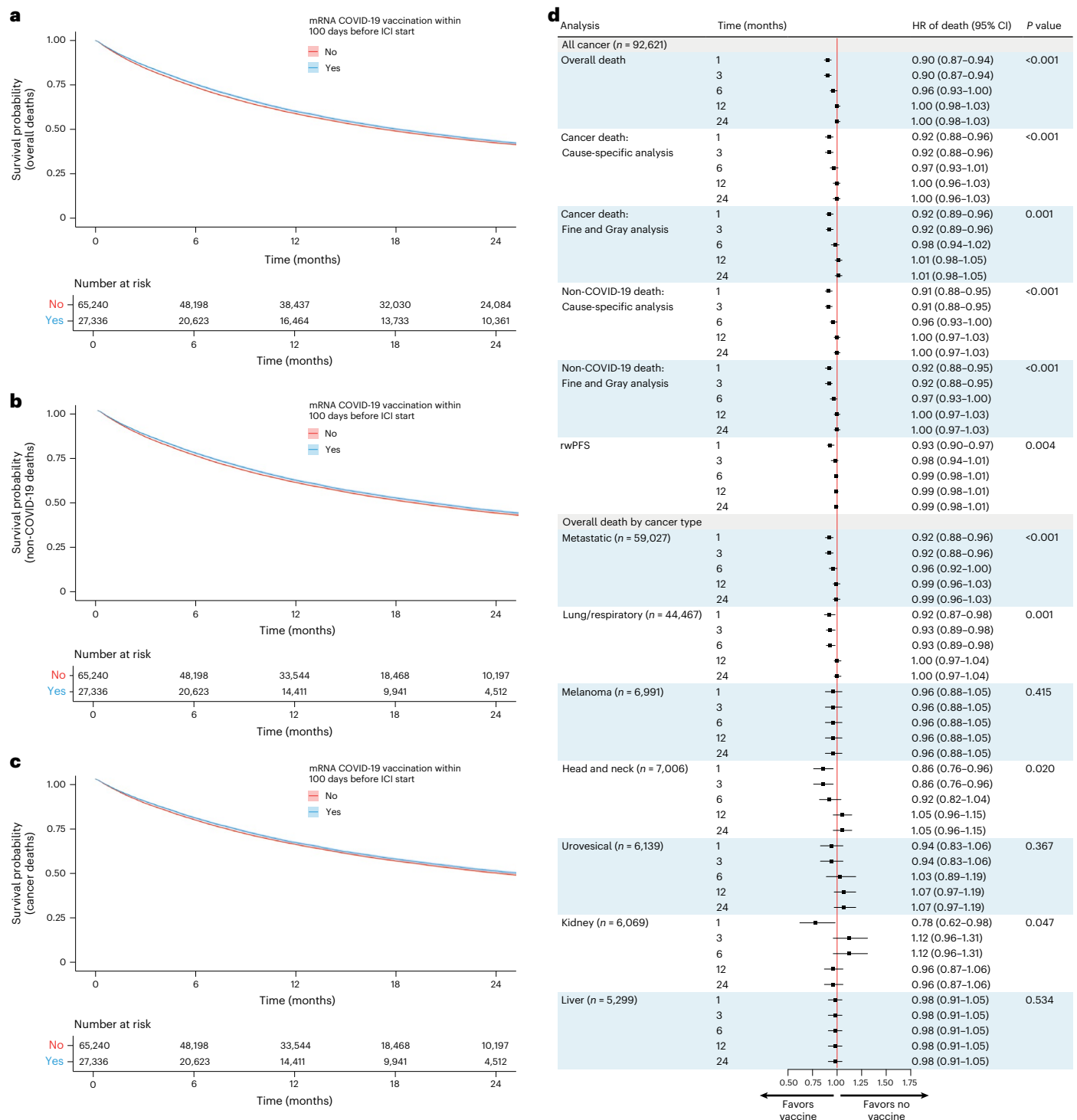


Fig. 3 | Association of pre-ICI mRNA COVID-19 vaccination (≥ 1 dose within 100 days before ICI initiation) versus no COVID-19 vaccination with survival.

a–c, Weighted survival curves are shown for overall death (**a**), non-COVID-19 death accounting for COVID-19 death as a competing event (**b**) and cancer death accounting for non-cancer death as a competing event (**c**). Numbers at risk shown beneath the curves in **a–c** correspond to weighted patient counts. Lines correspond to model-estimated survival rates and shaded bands to 95% CIs. **d**, The forest plot of the primary and secondary analyses, with the number

of included patients before weighting shown for each analysis. A total of 58,723 deaths, 48,580 non-COVID-19 deaths and 42,285 cancer deaths were observed during follow-up. Points represent HRs. Horizontal lines represent 95% CI, and the vertical line corresponds to an HR of 1. Robust score tests were used to derive *P* values. *P* value for overall death, 1.18×10^{-6} ; for cancer death (cause-specific analysis), 5.35×10^{-4} ; for non-COVID death (cause-specific analysis), 1.82×10^{-5} ; for non-COVID death (Fine and Gray analysis), 3.01×10^{-5} ; and for overall death in metastatic cancer, 3.29×10^{-4} .

the association was not driven by COVID-19-related mortality. Similar early associations were observed for cancer-specific mortality and real-world progression-free survival (rwPFS) (defined from ICI discontinuation or death)¹⁶, with no significant association thereafter,

consistent with the overall survival results (Fig. 3c,d). Likewise, analyses comparing non-mRNA COVID-19 vaccination with no COVID-19 vaccination in a 100-day pre-ICI period also showed beneficial effects on survival of vaccination, although with wider CIs because of smaller

numbers (Fig. 4). Similar estimates were observed when comparing in a 100-day pre-ICI period pneumococcal vaccination with no vaccination, as well as any vaccination (other than COVID-19) versus no vaccination, in analyses restricted to patients who had not received a COVID-19 vaccine in the pre-ICI period ($n = 65,226$) (Fig. 4 and Supplementary Fig. 5). By contrast, COVID-19 infection during a 100-day pre-ICI period was not associated with a significant increased risk of death (Fig. 4 and Supplementary Fig. 5). We also assessed potential residual confounding using negative-control outcomes, that is, evaluating whether the occurrence of acute cholecystitis, pulmonary embolism, femoral neck fracture, dentist visit and cataract surgery differed between individuals who were COVID-19 mRNA vaccinated in the pre-ICI period and those who were not; no significant difference was detected (Fig. 4). By contrast, the positive-control outcome analysis (COVID-19-related hospitalization) showed, as expected, a time-dependent protective effect of COVID-19 mRNA vaccine.

Post-ICI COVID-19 vaccination and overall survival

We then emulated a target trial to estimate the effect of receiving 1 or more doses of a COVID-19 mRNA vaccine within 100 days after ICI initiation (post-ICI period), compared with remaining unvaccinated against COVID-19, on all-cause mortality. This analysis was restricted to the 65,226 patients with no COVID-19 vaccination in the 100 days preceding ICI initiation (Fig. 2c). A CCW approach was used to address time-dependent vaccination and immortal-time bias¹⁷. After weighting, all covariates were well balanced between groups, with absolute standardized mean differences <0.10 , except for calendar year of ICI initiation and prior non-COVID-19 vaccination in the last year before the pre-ICI period (Supplementary Fig. 3). These additional analyses likewise suggested improved overall survival among patients who received an mRNA COVID-19 vaccine after ICI initiation compared with those who remained unvaccinated, with a similar time-dependent pattern to the pre-ICI mRNA analyses. Weighted median overall survival was 21.5 months (95% CI, 20.7–22.5) in the post-ICI mRNA-vaccinated group versus 19.0 months (95% CI, 18.6–19.5) in the unvaccinated group. In weighted time-varying models, post-ICI mRNA vaccination was associated with lower early all-cause mortality (HR, 0.84 (0.80–0.88) at 3 months and 0.86 (0.79–0.94) at 6 months), with no significant association at 12 months and thereafter (HR, 0.98 (0.93–1.04); Fig. 5a–d). *E* values were modest for these associations: 1.51 (1.41) and 1.46 (1.26) at 3 and 6 months, respectively. At 6 months after ICI initiation, the estimated survival probabilities were 75.4% (74.8–76.0) in the post-ICI mRNA-vaccinated group and 72.5% (72.1–72.9) among unvaccinated patients, corresponding to an absolute survival difference of 2.9% (2.3–3.5) (Supplementary Table 9). At 1 year, survival probabilities were 60.6% (59.8–61.4) and 57.2% (56.7–57.7), with an absolute survival difference of 3.4% (2.5–4.2); at 2 years, survival probabilities were 43.9% (43.0–44.7) and 41.3% (40.8–41.8), respectively, with an absolute survival difference of 2.6% (1.6–3.5). In weighted Cox models allowing for time-dependent effects, post-ICI mRNA vaccination was associated with an improved overall survival during the first months after ICI initiation, with similar results within the main cancer subtypes, and in the subgroup of patients with metastatic disease (Fig. 5d and Supplementary Fig. 6). Analyses restricted to non-COVID-19 deaths, treating COVID-19 death as a competing event, yielded very similar estimates (Fig. 5b,d). Similar early time-dependent associations were

observed for cancer-specific mortality and rwPFS, with disappearance of significant association over time, consistent with all-cause mortality results (Fig. 5c,d), suggesting that the observed association was not driven solely by COVID-19-related mortality. We also conducted subgroup analyses stratified by pre-ICI mRNA COVID-19 vaccination status to describe whether the post-ICI association differed according to vaccination status in the 100 days before ICI initiation. This analysis was restricted to patients initiating ICI in 2020–2021 ($n = 36,482$), because few patients received mRNA COVID-19 vaccines during both the pre- and post-ICI windows in later years. The mortality reduction was more pronounced among patients without pre-ICI mRNA vaccination (HR at 3 months, 0.82 (0.77–0.86); HR at 6 months, 0.81 (0.73–0.89)). By contrast, among patients who had received 1 or more pre-ICI mRNA dose, an association was observed only at 3 months (HR, 0.87 (0.81–0.94)); estimates attenuated and crossed the null at later time points (Supplementary Fig. 7). *E* values were modest for these associations: 1.56 (1.46), 1.58 (1.39) and 1.44 (1.26) for the corresponding point estimates.

Post-hoc sensitivity analyses accounting for early ICI discontinuation showed results broadly consistent with the main post-ICI analysis (Supplementary Fig. 8).

Discussion

In this nationwide administrative database with central recording of COVID-19 vaccination, in a cohort of more than 95,000 adults starting ICI therapy, pre- and post-ICI COVID-19 vaccinations were consistently associated with lower early all-cause mortality, with similar patterns for cancer-specific mortality. For mRNA COVID-19 vaccination in the pre-ICI window, the association was clearly time dependent (non-proportional hazards): the risk of death was lower during the first months after ICI initiation (for example, HR of 0.90 at 1–3 months and 0.96 at 6 months), with attenuation toward the null by about 1 year (HR ≈ 1 beyond 12 months). Accordingly, weighted survival curves separated early and then evolved largely in parallel, translating into modest absolute differences in survival probabilities (~ 1 –1.5% at 6–24 months) that likely reflect an early shift rather than a sustained reduction in risk over the entire follow-up. *E* values were modest, indicating that relatively weak unmeasured confounding could explain these associations; therefore, the 6-month and later associations should be considered statistically detectable but causally uninterpretable, particularly given their vulnerability to healthy-vaccinee and care-engagement confounding. Similar early patterns were observed when vaccination occurred after ICI initiation. Consistent findings were also observed in secondary analysis, including cancer death, non-COVID-19 deaths and rwPFS, as well as in sensitivity analyses using a shorter pre-ICI exposure window. Notably, comparable survival patterns were also seen with other vaccines in secondary analyses (for example, non-mRNA COVID-19 and pneumococcal vaccination), suggesting that the observed association may not be specific to COVID-19 mRNA vaccination. Cancer site-stratified analyses suggested broadly similar associations across major tumor types; nonetheless, limited sample sizes in some strata translated into wider confidence intervals and reduced power to detect modest differences between tumor sites.

Our findings extend to the population level ($\sim 95,000$ patients) signals from a recent clinical and biological single-center study, suggesting that COVID-19 mRNA vaccination may modulate ICI outcomes⁸;

Fig. 4 | Secondary and sensitivity analyses, including negative-control and positive-control outcomes: associations of pre-ICI exposures with overall mortality. Exposures were assessed in the 100 days before ICI initiation (50-day window, where indicated): COVID-19 vaccination (mRNA and non-mRNA), pneumococcal vaccination, any non-COVID-19 vaccination and documented COVID-19 infection. Vaccine analyses compare vaccinated with not vaccinated in the same window; infection analyses (restricted to unvaccinated patients) compare documented infection with none. The 50-day mRNA analysis was

restricted to ICI starts from 28 December 2020 to 11 May 2023. Sample sizes are before weighting. The number of included patients before weighting is shown for each analysis. Points represent HRs. Horizontal lines represent 95% CI, and the red vertical line corresponds to an HR of 1. Score tests were used to derive *P* values. *P* value for overall death for mRNA COVID-19 vaccination within 50 days versus unvaccinated, 8.53×10^{-11} ; for pneumococcal vaccination versus unvaccinated, 4.76×10^{-6} ; for any non-COVID-19 vaccination versus unvaccinated, 7.53×10^{-7} ; and for COVID-19-related hospitalization, 4.48×10^{-4} .



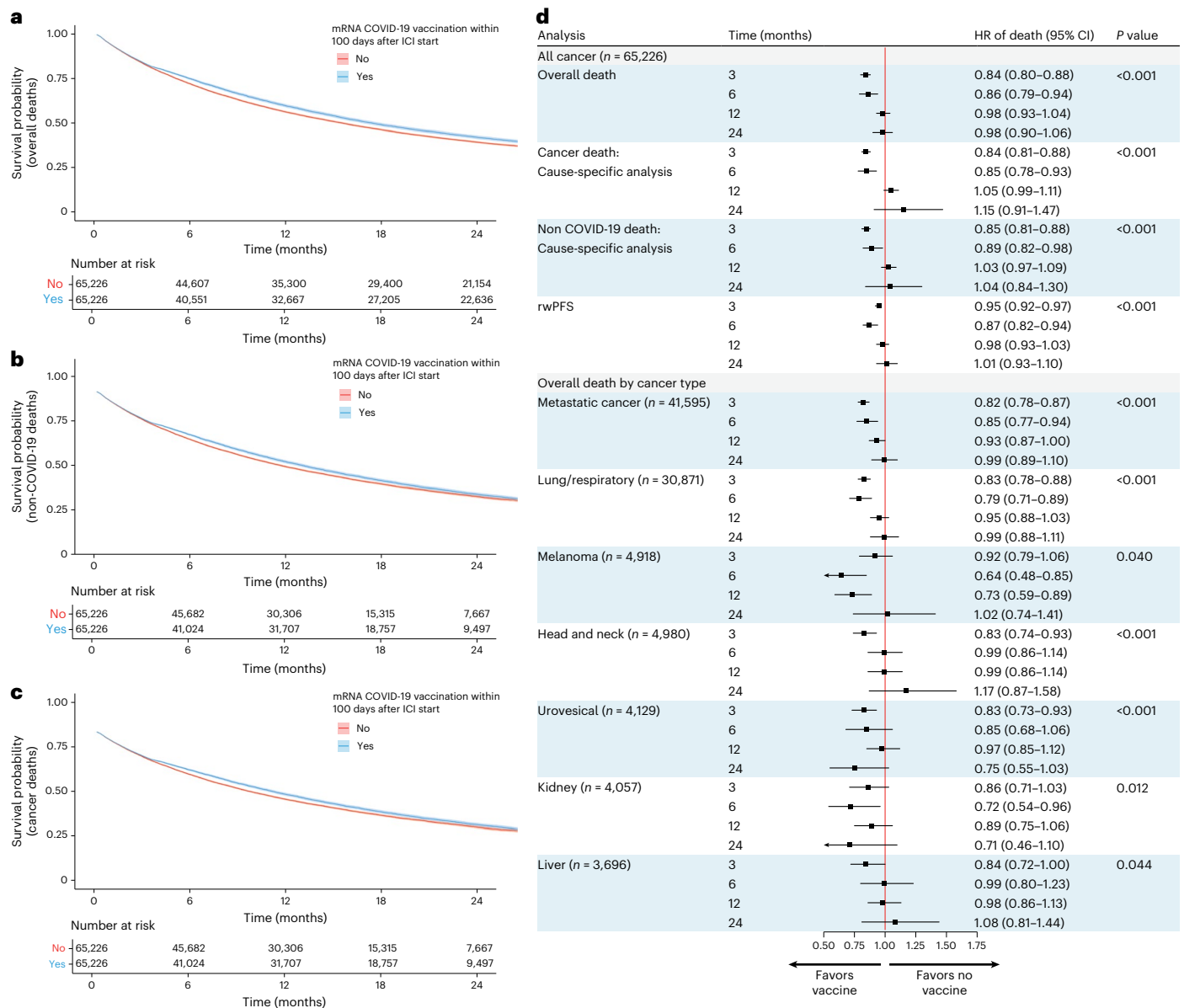


Fig. 5 | Association of post-ICI mRNA COVID-19 vaccination (≥ 1 dose within 100 days after ICI initiation) versus no COVID-19 vaccination with survival. **a–c, Weighted survival curves are shown for overall death (**a**), non-COVID-19 death treating COVID-19 death as a competing event (**b**) and cancer death accounting for non-cancer death as a competing event (**c**). **d**, HRs (points) with 95% CIs (horizontal lines) from the primary and secondary (cancer site) analyses of post-ICI mRNA COVID-19 vaccination. Under the CCW approach, each eligible**

patient is cloned at baseline into both strategies; numbers at risk therefore start at $n = 65,226$ in each arm. A total of 40,723 deaths, 33,373 non-COVID-19 deaths and 29,058 cancer deaths were observed during follow-up. P value for overall death, 1.22×10^{-36} ; for cancer death, 1.26×10^{-34} ; for non-COVID death, 1.03×10^{-33} ; for rwPFS, 1.55×10^{-30} ; and for overall death in metastatic cancer, 2.20×10^{-32} , in lung/respiratory cancer, 1.95×10^{-18} , in head and neck cancer, 2.41×10^{-4} , and in urovesical cancer, 3.67×10^{-4} .

however, in our data, the association was modest in magnitude and largely confined to the early months after ICI initiation. In their study, Grippin et al.⁸ reported that COVID-19 mRNA vaccination can ‘reset’ the tumor immune microenvironment through type I interferon-driven myeloid and T cell activation, potentially sensitizing otherwise poorly immunogenic tumors to PD-1/PD-L1 blockade; they also described a sizable early survival association with peri-ICI COVID-19 mRNA vaccination in 2 hospital-based cohorts (884 patients with stage III/IV non-small cell lung cancer and 210 with metastatic melanoma).

Several mechanisms could contribute to the observed survival association. First, COVID-19 vaccination reduces the risk of severe and fatal COVID-19, which has been associated with marked excess mortality and treatment disruption in patients with cancer^{4,18}. However, in our study, the associations between peri-ICI vaccination and survival

were similar when analyses were restricted to non-COVID-19 mortality, and early association was also observed for cancer-specific mortality. Together, these findings suggest that prevention of fatal COVID-19 alone is unlikely to fully explain the observed associations. Second, our results may partly reflect non-specific benefits and confounding factors that are not unique to COVID-19 vaccination. In the general French adult population aged 18–59 years, mRNA COVID-19 vaccination has been associated with lower long-term all-cause mortality, beyond the reduction in COVID-19-related deaths, an effect that is widely interpreted as a composite of direct protection, healthier baseline status among vaccine recipients and broader healthcare engagement¹⁸. In our ICI cohort, we observed similar directions and magnitudes of association, and we found a concordant survival beneficial association when comparing non-mRNA COVID-19 vaccines or other adult vaccines (such

as influenza or pneumococcal) with no vaccination. These concordant cross-vaccine patterns may reflect either genuine non-antigen-specific immune effects, such as bystander non-specific immune stimulation, or non-causal correlates of vaccine uptake, including healthy-vaccinee selection and care engagement. These explanations have opposite implications for the Grippin et al.⁸ hypothesis, and our observational design cannot adjudicate between them. A healthy-vaccinee and care-engagement mechanism remains a plausible explanation of this association, because vaccination requires patients to remain alive, clinically stable and connected to care. Although adjustment for a large number of potential confounders, control exposures and control outcomes reduces some concerns, these approaches cannot exclude residual confounding, in particular by evolution of performance status, tumor burden, disease trajectory or clinician-assessed prognosis⁹. A biological interaction between vaccination-related immune activation and ICI remains possible, and the larger early association observed for post-ICI vaccination is compatible with this hypothesis. However, this pattern is also compatible with stronger healthy-vaccinee and care-engagement bias during the post-ICI grace period, and the broadly similar associations observed with non-mRNA COVID-19 vaccines, pneumococcal vaccination and other adult vaccines prevent attributing the survival pattern to an mRNA-specific mechanism^{6–8}. Notably, we did not observe a comparable survival beneficial association in analyses with documented COVID-19 infection as exposure in the same pre-ICI window, which makes a purely infection-driven stimulation explanation less likely. However, this null finding should be interpreted cautiously, because potential immune stimulation from infection could have been offset by competing harms, including severe illness, treatment disruption or COVID-19-related mortality.

From a clinical standpoint, the key message is that COVID-19 vaccination around the time of ICI initiation was not associated with worse survival in this large real-world population. This finding should reassure clinicians and patients who may worry that vaccination could blunt ICI efficacy or exacerbate immune-related life-threatening toxicities, concerns that have been fueled by case reports and small series^{4,5}. Although vaccinated patients had better survival overall, this association was modest, transient and not specific to COVID-19 mRNA vaccines, which limits causal interpretation. Accordingly, our results are best interpreted as reassuring evidence that peri-ICI vaccination does not seem to compromise outcomes and as being consistent with current recommendations to prioritize COVID-19 vaccination in patients with cancer receiving ICI, rather than as evidence that vaccination enhances ICI efficacy or improves cancer outcomes^{11,12}.

Several limitations should be acknowledged. First, as in any observational study, residual confounding cannot be fully excluded. Although we adjusted for a broad range of sociodemographic, oncologic, comorbidity, frailty and healthcare-use variables, the SNDS does not capture some key clinical factors such as direct measures of performance status, PD-L1 expression, detailed tumor burden and certain life-style characteristics. In addition, vaccination in the post-ICI period may partly reflect preserved functional status and continued healthcare engagement; thus, patients with rapidly progressive disease may have been less likely to be vaccinated after ICI initiation despite the CCW and IPTW approaches. Residual imbalance in calendar year and prior non-COVID-19 vaccination persisted after weighting in the post-ICI analysis. Calendar time may have biased estimates through changes in vaccine uptake, COVID-19 epidemiology, ICI practice and outcomes, although the direction of bias is difficult to predict. This concern was partly addressed by the 2020–2021 restricted analysis, in which all variables were well balanced after weighting and results showed a similar early pattern (Supplementary Fig. 7). The SNDS-derived poor-performance status proxy should not be interpreted as a direct measure of Eastern Cooperative Oncology Group Performance Status (PS). However, its prevalence was 19.3% overall and 23.9% in metastatic cancers, broadly consistent with estimates from real-world ICI-treated

populations^{19–21}. Indeed, Eastern Cooperative Oncology Group PS ≥ 2 influences treatment selection in routine practice, and patients with Eastern Cooperative Oncology Group PS 3–4 are expected to represent only a limited proportion of patients who actually initiate ICI^{19–21}. The absence of association with negative-control outcomes provides only limited reassurance, because these outcomes may not fully capture the performance status and tumor-trajectory gradients most relevant to cancer mortality. An ideal negative-control outcome strongly linked to these gradients, but unrelated to vaccination, was not available in the SNDS. Second, our exposure definitions relied on prespecified 100-day peri-ICI windows to further investigate the findings of Grippin et al.⁸, and most vaccinated patients received only 1 dose within these windows. This limited our ability to assess dose response, dose sequence or precise timing, thereby limiting interpretation in terms of antigen-specific immune priming. Third, interpretation of secondary endpoints requires caution. Cause-of-death information was only available through 2023, resulting in a shorter inclusion period and follow-up for cause-specific analyses than for all-cause mortality. Moreover, cancer-specific mortality derived from administrative data should not be interpreted as direct evidence of improved antitumor efficacy, given the possibility of misclassification. In addition, rwPFS was based on ICI discontinuation or death and should not be interpreted as a direct measure of antitumor efficacy, because SNDS does not reliably distinguish progression-related discontinuation from toxicity, planned treatment stop, or changes in care pathway. Fourth, some secondary analyses—notably those involving non-mRNA COVID-19 vaccines and certain cancer types—were based on relatively small numbers, leading to wide CIs and reduced power to detect modest differences. Finally, our findings reflect vaccination practices, circulating variants and oncologic care patterns in France during 2020–2023 and may not fully generalize to other healthcare settings, later pandemic phases or newer vaccine formulations.

In a large, nationwide cohort of patients with cancer initiating ICI therapy, peri-ICI COVID-19 vaccination was associated with a modest early survival association. The association persisted when restricting to non-COVID-19 deaths, and similar early patterns were observed for cancer-specific outcomes, although these findings do not establish a direct effect on antitumor efficacy. These latter patterns were also observed for non-mRNA COVID-19 vaccines and for other adult vaccinations, arguing that the observed association may not be specific to mRNA COVID-19 vaccination and may rather reflect non-specific correlates of vaccination and/or residual confounding. Overall, these findings are reassuring regarding the use of COVID-19 vaccination in ICI-treated patients. Only prospective, randomized and mechanistically oriented studies would allow for a definitive determination of whether there is a causal link between vaccination and mortality in the context of initiating ICI therapy.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-026-04640-y>.

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Methods

This study is reported in accordance with the STROBE (Statement: guidelines for reporting observational studies) and RECORD-PE (REporting of studies Conducted using Observational Routinely-collected Data-Pharmaco-Epidemiology) guidelines (see RECORD-PE checklist document).

Data source

We conducted a nationwide observational study using the French National Health Data System (SNDS), including all patients with cancer initiating ICIs in France, to assess the association between COVID-19 vaccination around ICI initiation and overall mortality. R.C. and A.A. have permanent access to the SNDS for this research under articles R.1461-12 et seq. of the French Public Health Code and the French Data Protection Authority decision (CNIL-2016-316). The SNDS contains individual-level pseudonymized information on nearly all residents (~68 million) covered by the French statutory health insurance schemes. It links three main national datasets through a unique lifelong patient identifier. The national health insurance claims database (Datamart de Consommation Inter-Régime, DCIR) provides demographic characteristics, reimbursed outpatient drugs and procedures, and long-term disease status. The PMSI database contains hospital discharge diagnoses, procedures and costly drugs administered during hospitalization. The French causes of death registry provides dates and cause of death, with data available up to 31 December 2023. Exhaustive information on COVID-19 vaccination (vaccine type and brand, and date of each dose) and COVID-19 tests (negative and positive RT-PCR or antigen tests) is recorded in additional dedicated national modules available in the SNDS up to 30 June 2024 and 30 June 2023, respectively. Detailed code lists used to define all variables are provided in the Supplementary Appendix (for example, ICD-10, *International Statistical Classification of Diseases and Related Health Problems*, 10th revision).

Study population

All adults (aged ≥ 18 years) with a cancer diagnosis (see Supplementary Appendix, eTable-1, for cancer types coding details) who initiated ICI therapy between 28 December 2020 (first availability of COVID-19 vaccines in France) and 22 March 2023 (100 days before the end of available COVID-19 infection data in the SNDS) were eligible. ICI initiation (time 0) was defined as the date of the first in-hospital administration of an anti-PD-1, anti-PD-L1 or anti-CTLA-4 agent. Patients were required to have at least 3 years of observable history in the SNDS before ICI initiation to allow ascertainment of baseline covariates. Patients with prior solid organ or bone marrow transplantation or living with human immunodeficiency virus infection were excluded (see Supplementary Appendix, eTable-2, for codes).

Study overview

We used causal inference methods applied to the French SNDS to estimate the effect of peri-ICI COVID-19 mRNA vaccination (BNT162b2 or mRNA-1273) on all-cause mortality among adult patients with cancer initiating ICI. Because only about 5% of patients received a COVID-19 mRNA vaccine in both the 100 days before and the 100 days after ICI initiation (almost all in 2021), we analyzed 2 peri-treatment windows separately: a pre-ICI period, defined as vaccination within 100 days before ICI initiation, and a post-ICI period, defined as vaccination within 100 days after ICI initiation. For the pre-ICI period, we compared COVID-19 mRNA vaccinated and unvaccinated patients using IPTW to address baseline confounding (see Supplementary Appendix eTable-3 for details). For the post-ICI period, we emulated a target trial to compare strategies of receiving at least 1 dose of COVID-19 mRNA vaccine versus not receiving any COVID-19 vaccine in the 100-day period after ICI initiation, defined as the grace period (see Supplementary Appendix eTable-4 for details). In this analysis, ‘pre-ICI unvaccinated’ refers only

to the absence of COVID-19 vaccination during the 100-day pre-ICI window, not to vaccine-naïve status. This target-trial emulation framework makes the causal questions explicit and helps limit immortal-time and other design-related biases^{22–24}. Individuals who received mixed or only non-mRNA COVID-19 vaccines during each analysis window were (1) excluded from the pre-ICI initiation analysis or (2) censored in the post-ICI initiation analysis, respectively.

Outcomes

The primary outcome was all-cause mortality. Date of death was directly available in the SNDS through the French National Institute of Statistics and Economic Studies (INSEE).

Secondary outcomes included (1) non-COVID-19 mortality, defined as death with an underlying cause coded other than ICD-10 U07.1 (‘COVID-19, virus identified/laboratory confirmed’) or ICD-10 U07.2 (‘COVID-19, virus not identified/clinically or epidemiologically diagnosed’), in accordance with WHO guidance^{25,26}; (2) cancer mortality, defined as deaths with an underlying cancer cause²⁷; and (3) rwPFS. rwPFS was defined as the time to ICI discontinuation or death, with ICI discontinuation being defined by a ≥ 81 -day interval without ICI administration (21-day theoretical treatment coverage and an authorized gap of 2 months)¹⁶. Analyses of cause-specific mortality were restricted to deaths occurring through the end of 2023, as data are not yet available for 2024.

Covariates

Baseline covariates included in the models are shown in Table 1. They included age (categories), sex, low-income status, deprivation index (socioeconomic deprivation score)²⁸ and region of residence. Cancer-related variables included the primary tumor site (categorized as lung/respiratory, melanoma, head and neck, urovesical, kidney, liver, gastrointestinal, mesothelioma/pleural and other), a proxy for metastatic disease (presence of metastasis codes), the type of ICI regimen (anti-PD-1, anti-PD-L1 or anti-PD-1+anti-CTLA-4), cancer radiotherapy, chemotherapy or major cancer surgeries in the last 3 years and inpatient (versus outpatient) initiation of ICI as a proxy of a poorer performance status²⁹. Centers where ICI was initiated were classified according to patient volume as high volume (top 10% of centers), intermediate volume (10–40%) and low volume (bottom 60%)²⁹. Cancer-related frailty variables were assessed during the 6 months before ICI initiation and updated during the 100 days after initiation. These included malnutrition, at least two ambulance transports on separate days, at least one emergency department visit, at least one intensive care unit admission, at least one hospitalization in a rehabilitation or long-term care facility, at least one at-home hospitalization, at least one home nursing care visit, use of home oxygen therapy and at least one palliative care stay. Baseline treatments before ICI and time-updated treatments during the 100 days after initiation included anti-angiogenic agents, opioids, antibiotics, antiepileptics, bisphosphonates, anti-receptor activator of nuclear factor kappa-B ligand agents, proton pump inhibitors and corticosteroids (defined as at least 7 mg per day of prednisone). We also included the number of general practitioner consultations in the previous 6 months as an additional proxy for health-seeking behavior and receipt of any non-COVID vaccination within 1 year before ICI initiation as a proxy for vaccination-related behavior. Other comorbidities identified in the past medical history included serious chronic respiratory disease (respiratory failure, chronic obstructive pulmonary disease or emphysema), heart failure, hypertension, diabetes and obesity in the past 2 years. Calendar year of ICI initiation was used to account for changes in COVID-19 epidemiology, vaccine availability and oncologic practice over time. Finally, COVID-19 infection before and after ICI initiation was defined as a recorded positive test or a hospitalization with a COVID-19 diagnosis code in the available follow-up (see Supplementary Appendix eTable-2 for codes).

Statistical analysis

Categorical data were reported as number (%). Quantitative data were reported as median and IQR. Missing values were imputed using the median for ordinal variables and the mode for categorical variables.

In the pre-ICI period, we estimated the effect of receiving a COVID-19 mRNA vaccine versus no vaccine on all-cause mortality using a Cox proportional hazards model weighted by IPTW. Weights were estimated using a logistic regression model including baseline covariates, and stabilized weights were derived to reduce variance (Supplementary Fig. 2).

In the post-ICI window, we emulated a trial comparing dynamic strategies of vaccination within 100 days after ICI start with no vaccination in that period among patients not vaccinated in the pre-ICI period. A CCW approach¹⁷ was implemented to address confounding and immortal-time bias during the 100 days after ICI initiation. Each eligible individual was cloned into two copies, one assigned to each strategy. Follow-up for each clone began at ICI initiation. In the 'vaccination' strategy, a clone was censored if the individual had not received the relevant vaccine by 100 days after ICI initiation; in the 'no vaccination' strategy, a clone was censored at the time of the first mRNA COVID-19 vaccination. In both strategies, clones were additionally censored at the time of any non-mRNA COVID-19 vaccination occurring within the first 100 days after ICI start. Follow-up ended at death or at the end of data availability. Deaths occurring during the 100-day grace period were counted as events in the strategy arms with which they were compatible, up to death, that is, in both arms if death occurred without being vaccinated before and in the vaccination arm only if they died after being vaccinated. Artificial censoring and inverse probability of censoring weighting were applied only for deviation from the assigned vaccination strategy. To adjust for baseline and time-varying confounding and for informative censoring induced by this artificial censoring, we estimated inverse probability of censoring weighting using Cox regression models including baseline covariates and, where applicable, time-varying variables such as COVID-19 infection within the first 100 days after ICI start (Supplementary Fig. 3 for full list of variables included).

In the two analyses (IPTW and CCW), covariate balance before and after weighting was assessed by calculating standardized mean differences; absolute values <0.1 were considered indicative of adequate balance. Weighted Kaplan–Meier estimators were used to estimate survival curves by exposure group, and weighted Cox proportional hazards models were fitted. Robust variance estimators accounted for clustering of clones within individuals in the CCW approach. Proportional hazards assumptions were assessed by scaled Schoenfeld residuals. When this assumption was not met, we modeled time-varying effects by splitting the follow-up into intervals in which the proportional hazards assumption was verified and held, and by including interactions between exposure and time such as t or $\log(t)$, when appropriate, to obtain time-varying HRs at clinically relevant time points. Otherwise, we reported a single HR, considering the effect was not time dependent.

We reported effect estimates primarily as HRs with 95% CIs and complemented with absolute differences in survival probabilities at 1 (for the pre-ICI analyses only), 3, 6, 12 and 24 months and 95% CI derived using bootstrap resampling on the weighted sample (1,000 replicates for both the pre-ICI and post-ICI analyses). We calculated E values for the HRs quantifying the association between mRNA COVID-19 vaccination and mortality, and the 95% confidence limits closest to the null to quantify sensitivity to residual unmeasured confounding. For protective associations, HR were inverted before applying the standard E -value formula. Lower E values indicate that weaker unmeasured confounding could fully explain the observed association.

Secondary analyses were main analysis stratified by cancer type with at least 5,000 patients in all analyses (metastatic, lung/respiratory, melanoma, head and neck, urovesical, kidney and liver cancers), and main analysis with (1) non-COVID-19 mortality as the outcome (and COVID-19 death as competing risk), (2) cancer mortality as the

outcome (and non-cancer death as competing risk) and (3) rwPFS as the outcome. Cause-specific Cox models were used throughout as the primary competing-risk approach because our aim was to estimate the direct effect of vaccination on the event of interest, not mediated by competing events. Fine and Gray models were applied as sensitivity analyses only in the pre-ICI IPTW setting, where exposure was fixed at baseline, and not in the post-ICI CCW analysis because the latter analysis uses time-dependent exposure and covariate integration, which is not compatible with the Fine and Gray models³⁰.

Sensitivity analyses (in the pre-ICI period) examined the association between overall death and (1) non-mRNA COVID-19 vaccines (restricted to 2021), (2) pneumococcal vaccination (among COVID-19 unvaccinated individuals), (3) any vaccination (excluding COVID-19 vaccines, among COVID-19 unvaccinated individuals) and (4) recent COVID-19 infection (among COVID-19 unvaccinated individuals). We also tested alternative exposure windows of 50 days, negative-control outcome analysis using acute cholecystitis, pulmonary embolism, femoral neck fracture, dentist visit and cataract surgery, and positive-control outcome analysis using COVID-19-related hospitalization to investigate residual confounding.

Post-ICI sensitivity analyses accounted in addition for early ICI discontinuation, as a proxy of rapid clinical deterioration, early progression, treatment changes or radiographic progression. First, among individuals with no COVID-19 vaccination in the pre-ICI period who were alive and still receiving ICI 100 days after initiation, we compared receipt of at least 1 dose of COVID-19 mRNA vaccine within the first 100 days after ICI initiation with no COVID-19 vaccination during the same period using ICI initiation + 100 days as index date. Second, we performed an additional sensitivity analysis within the CCW target-trial emulation analysis, and censoring (and weighting) at early ICI discontinuation also. In the latter analysis, the strategies required both adherence to the assigned vaccination strategy and continuous ICI treatment through day 100. Accordingly, clones were censored at early ICI discontinuation during the first 100 days, in addition to censoring at deviation from the assigned vaccination strategy, and inverse probability of censoring weights were recalculated to account for this additional censoring mechanism.

In the SNDS, loss to follow-up for all-cause mortality is expected to be very low because the national claims database covers nearly all individuals residing in France, serves as the basis for healthcare reimbursement and records dates of death through linkage to the national death register. However, loss to follow-up cannot be assumed to be zero, for example, in cases of emigration or administrative linkage failure.

Data management, including cohort construction, was performed using SAS Enterprise Guide version 4.3. All statistical analyses, including weighting, survival models, bootstrap estimation of absolute survival differences and figure generation, were performed using R version 4.2.1. Two-sided $P < 0.05$ was considered statistically significant; no adjustment for multiple testing was done, as secondary, subgroup and sensitivity analyses were considered exploratory.

Ethics

The study used pseudonymized data from the SNDS and complied with French regulations on the secondary use of health data. Use of SNDS data for research purposes is approved by the relevant French data protection authority and ethics committee, with a waiver of informed consent granted because the study relied solely on routinely collected administrative data. Individuals covered by the health insurance have been informed that their data could be re-used for research purposes and have the rights of access, rectification, opposition to reuse and erasure of their data.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The SNDS contains third-party, pseudonymized individual-level data that are not publicly available. In accordance with French data protection regulations, access is granted only to authorized investigators for projects serving the public interest, following the relevant regulatory approvals, and analyses must be conducted within an approved secure environment; only aggregate, non-identifying results may be exported. Researchers wishing to access the SNDS data can submit a request through the Health Data Hub (*Plateforme des données de santé*) in line with applicable governance procedures.

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Author contributions

J.-E.S., J.C. and F.T. conceived and supervised the study. J.-E.S., R.C., A.A. and F.T. designed the study. R.C. and A.A. accessed the SNDS data and performed cohort construction and statistical analyses, with methodological input from Y.D.R. J.-E.S., R.C., A.A., Y.D.R., J.C. and F.T. interpreted the results. J.-E.S. drafted the paper. All authors critically revised the paper, approved the final version and agreed to be accountable for the work.

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Competing interests

The authors declare no competing interests.

Additional information

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Software and code

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Data collection	No primary data were collected for this study. We used routinely collected, pseudonymized administrative healthcare data from the French National Health Data System (SNDS). Data extraction, linkage and dataset preparation were conducted within the secure SNDS environment by authorized analysts using the standard tools provided by the data custodian (e.g., SQL queries and SAS programs). No bespoke software was used specifically for data collection.
Data analysis	All statistical analyses were performed using SAS Enterprise Guide / SAS (SAS Institute Inc., Cary, NC, USA; version 4.3) and R. version 4.2.1. Custom scripts were used for cohort construction (definitions of exposure, outcomes and covariates) and for the analyses (e.g., weighting procedures, Cox/Fine–Gray models, bootstrap resampling).

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The SNDS contains third-party, pseudonymized individual-level data that are not publicly available. In accordance with French data-protection regulations, access is granted only to authorized investigators for projects serving the public interest, following the relevant regulatory approvals, and analyses must be conducted within an approved secure environment; only aggregate, non-identifying results may be exported. Researchers wishing to access the SNDS data can submit a request through the Health Data Hub (Plateforme des données de santé) in line with applicable governance procedures.

Research involving human participants, their data, or biological material

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Reporting on sex and gender

Sex was available in the SNDS as an administrative variable (male/female) recorded for health insurance purposes. In the overall cohort (N=95,015), 61,945 (65.2%) were men and 33,070 (34.8%) were women. Sex was included as a baseline covariate in the weighting models (IPTW and CCW) to control for confounding. The study was not restricted to a single sex, and no sex-stratified analyses were prespecified. The SNDS does not capture gender identity/presentation; therefore, gender-based analyses were not performed.

Reporting on race, ethnicity, or other socially relevant groupings

Race/ethnicity variables are not collected in the SNDS and were therefore not available or used in this study. No proxy variables for race/ethnicity were used. Socioeconomic position was partially captured using low-income status and deprivation index, a territorial index of social to characterize the socioeconomic environment of beneficiaries based on their municipality of residence (as available in SNDS). These variables were included as covariates to control confounding, along with demographic characteristics, cancer-related variables, cancer-related frailty indicators, comorbidities, calendar year, proxies of health seeking behaviour and other clinical proxies.

Population characteristics

Study population and COVID-19 vaccination patterns

Between December 28, 2020 (first availability of COVID-19 vaccines in France) and March 22, 2023 (end of available COVID-19 infection reporting in France), we included 95,015 adults initiating ICI in the French national health data system (SNDS, Flow-chart in Figure-1; Table-1 for general characteristics). The median age [IQR] at ICI initiation was 67 years [60–74] and 65.2% were men. Lung and other respiratory cancers represented nearly half of the cohort (45,825; 48.2%), followed by head and neck (7,216; 7.6%), melanoma (7,147; 7.5%), kidney (6,257; 6.6%), urovesical (6,233; 6.6%), liver (5,495; 5.8%), gastrointestinal (4,372; 4.6%), breast (4,094; 4.3%), and the rest in other cancer sites (Figure-2A). Most patients received anti-PD-1 monotherapy (71.2%), with smaller proportions treated with anti-PD-L1 (23.5%) or anti-PD-1 plus anti-CTLA-4 (5.3%, Figure-2B). A total of 10.1% of patients received anti-angiogenic combined or within the 3 months prior to ICI initiation. Comorbidities were frequent, including hypertension (58.6%), diabetes (23.3%), chronic obstructive pulmonary disease/emphysema (18.5%), and heart failure (7.3%). The details per cancer types are displayed in Supplementary Tables 1-6.

The COVID-19 vaccination patterns in the peri-ICI period of interest are summarized in the Sankey diagram in Figure-2C, highlighting the large predominance of mRNA-only COVID-19 vaccination and the rarity of mixed or non-mRNA schedules. Within the 100 days before ICI initiation (termed pre-ICI period), 27,395 patients (28.8%) had received at least one dose of an mRNA COVID-19 vaccine only (median doses [IQR]: 1[1-1]), 2,326 (2.4%) had received at least one dose of non-mRNA COVID-19 vaccine only (median doses: 1[1-1]), 68 (0.1%) had received at least a dose of an mRNA COVID-19 vaccine and of a non-mRNA COVID-19 vaccine and 68.6% were not vaccinated for COVID-19 (Figure-2C). Uptake of mRNA vaccination before ICI initiation was between 31-35% in the 6 most prevalent cancer subtypes tested (Supplementary Tables 1-6 for details). In the analysis investigating the effect of COVID-19 vaccination within 100 days after ICI initiation (termed post-ICI period), we focused on patients who had received no COVID-19 vaccine in the pre-ICI period (n=65,226). Among them, 18,135 (27.8%) initiated a COVID-19 vaccine (n=17,514 mRNA only, 1[1-2] doses; n=602 non-mRNA, 1[1-1]dose; and 19 with at least one dose of both types) in the post-ICI period. In total, 47 091 received no COVID-19 vaccine in both the pre- and post-ICI periods. Among patients unvaccinated with a COVID-19 vaccine in the pre-ICI period, 2,176 (3.3%) received a pneumococcal vaccine and 9,856 (15.1%) received at least 1 dose of any non-COVID-19 vaccine. An upset plot showing vaccination overlaps in the pre-ICI period is presented in Supplementary Figure-1.

Recruitment

All adults (aged ≥18 years) with a cancer diagnosis (see Appendix eTable-1 for cancer types coding details) who initiated ICI therapy between December 28, 2020 (first availability of COVID-19 vaccines in France) and March 22, 2023 (100 days prior to the end of available COVID-19 infection data in the SNDS) were eligible. ICI initiation (time zero) was defined as the date of the first in-hospital administration of an anti-PD1, anti-PDL1 or anti-CTLA4 agent. Patients were required to have at least 3 years of observable history in the SNDS before ICI initiation to allow ascertainment of baseline covariates. Patients with prior solid organ or bone marrow transplantation or living with human immunodeficiency virus infection were excluded (see Appendix eTable-2 for codes).

Ethics oversight

The study used pseudonymized data from the SNDS and complied with French regulations on the secondary use of health data. Use of SNDS data for research purpose is approved by the relevant French data protection authority and ethics committee with a waiver of informed consent granted because the study relied solely on routinely collected administrative

data. Individuals covered by the health insurance have been informed that their data could be re-used for research purpose and have the rights of access, rectification, opposition to reuse, and erasure of their data.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

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Sample size	No formal sample-size calculation was performed because this was a nationwide, population-based study using the SNDS. We included all eligible adults with cancer initiating an immune checkpoint inhibitor in France during the prespecified study period (28 December 2020 to 22 March 2023). The resulting cohort size (N=95,015) reflects the underlying population and provided sufficient numbers of vaccinated and unvaccinated person-time to estimate relative risks with adequate precision.
Data exclusions	Exclusion criteria were predefined and applied before analyses according to the study protocol. Individuals were excluded if they did not meet eligibility criteria (e.g., age <18 years), lacked continuous health insurance coverage required to ascertain baseline covariates, or had missing/invalid key identifiers or dates precluding correct follow-up. No additional data were excluded after cohort entry beyond these prespecified criteria.
Replication	This study is based on secondary analysis of routinely collected administrative healthcare data and does not include experimental procedures requiring replication. Reproducibility was addressed through prespecified cohort definitions and analytic protocols, use of standardized SNDS algorithms, and internal code review and verification (including consistency checks and sensitivity analyses). Results were robust across the main and sensitivity analyses.
Randomization	Participants were not randomized. Exposure (COVID-19 vaccination) occurred in routine care. Confounding was addressed using causal-inference approaches, including inverse probability of treatment weighting (IPTW) and a target-trial emulation using cloning–censoring–weighting, with adjustment for a wide set of baseline covariates.
Blinding	Blinding was not applicable. The study used pseudonymized administrative data; investigators did not interact with participants and did not influence exposure or outcomes. Analyses were conducted on de-identified datasets within the secure SNDS environment.

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Methods

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Clinical data

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Clinical trial registration	Not applicable. This is a nationwide observational study based on routinely collected, pseudonymized administrative healthcare data (SNDS) and not a clinical trial. We thus completed the RECORD-PE checklist which is appropriate for our study instead of the CONSORT checklist)
Study protocol	The study protocol can be provided to editors/reviewers upon request. All relevant informations are also reported in the article as recorded in the RECORD-PE checklist in appendix in the article

Data collection	Data were obtained from the French National Health Data System (SNDS), a nationwide administrative healthcare database covering the French population, including outpatient reimbursements, hospital discharge diagnoses/procedures, and vaccination records.
Outcomes	Outcomes were prespecified in the protocol and are reported with codes used in appendix in the article. The primary outcome was all-cause mortality, with date of death obtained via linkage in the SNDS to the National Institute for Statistics and Economic Studies (INSEE). Secondary outcomes included (1) non-COVID-19 mortality, defined as death with an underlying cause coded other than ICD-10 U07.1 ("COVID-19, virus identified/laboratory confirmed") or ICD-10 U07.2 ("COVID-19, virus not identified/clinically or epidemiologically diagnosed"), in accordance with WHO guidance; (2) Cancer mortality, defined as deaths with an underlying cancer cause and (3) real-world Progression Free Survival.

Plants

Seed stocks	Not applicable
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Authentication	Not applicable